

On the Optimality of Attention-Allocated Retrieval: Computable Thresholds of Federated Retrieval Pipelines

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Abstract

Water-filling attention allocation—theorem T2 of the split-attention agent framework (Sachikonye, 2026b)—is optimal under a diminishing-returns assumption which that paper is careful to flag as a property of the *environment* rather than of the allocator. This paper measures the assumption in one environment that matters: a federated retrieval pipeline that must join records across endpoints before a reasoner can be invoked over them (Doerr and Born, 2025; Sachikonye, 2026c). The assumption is false, and it fails in a way that is exactly characterised.

Expected join yield after e queries is $Y(e) = Np^2$ with $p = 1 - (1 - 1/N)^{e/2}$, because a join requires *both* sides of a pair to have been drawn. Squaring a saturating function gives a sigmoid, so retrieval has *increasing* returns early. We show (i) the inflection is exact, $e^* = 2 \ln 2 / \ln(1/(1 - 1/N)) \rightarrow 2N \ln 2$, and the yield there is $N/4$ —*exactly a quarter of the joinable corpus for every N* , so the threshold is computable at run time from a count query with no fitted constant (Theorem 2.3); (ii) the threshold *reverses the optimal policy* rather than bounding lost optimality: below e^* the optimum is at a corner and committing the whole budget to one join beats spreading it by up to 97%, while above e^* the ordering reverses and the even split wins (Theorem 3.1); and (iii) the optimum equalises effort measured in units of each join’s *own* threshold, e/e^* , not raw effort—which is what makes a single shadow price meaningful across corpora of different sizes (Theorem 3.5).

We then build the allocator these results dictate and check it against exhaustive search over the simplex, with which it shares no reasoning. It agrees to within 0.0000% at two sources and 10^{-6} at three, and below threshold it *refuses*: it declines to allocate and names the budget at which it could, a refusal of the same shape as the two the medium-vertex paper derives from the medium (Sachikonye, 2026a), but with a threshold computable from corpus size rather than fitted. Twenty-one checks pass, six of them negative controls that must fail on well-formed input.

We report two of our own measurement errors and one genuine flaw in the allocator, because each is load-bearing. The first two made the finding look like an artefact—Monte Carlo noise five times the true curvature, then a parity artefact—and were settled only by evaluating the closed form with no sampling at all. The third is more instructive: budget $\geq \sum_i e_i^*$ is *necessary but not sufficient* for the interior solution to be optimal, and an allocator built on that false sufficiency loses 8% at the boundary. No self-consistency check could have caught it; only the brute-force comparison did.

Keywords: attention allocation; water-filling; diminishing returns; federated query; convexity threshold; corner solution; shadow price; refusal; negative control; knowledge graph retrieval.

Contents

1 Introduction

2

1.1	An assumption flagged and not tested	2
1.2	Why joins are the case that matters	2
1.3	What we prove	2
1.4	What we do not claim	3
2	The join yield surface	3
2.1	Model	3
2.2	The threshold	3
3	The threshold reverses the optimal policy	4
3.1	Corner versus interior	4
3.2	Normalised effort	6
4	The allocator	8
4.1	What the results dictate	8
4.2	Interface	8
4.3	A necessary condition mistaken for a sufficient one	9
5	Validation	9
5.1	What the suite can establish, and what it cannot	9
5.2	Results	11
5.3	Two measurement errors and two real ones	11
6	Scope	14
7	Conclusion	14

1 Introduction

1.1 An assumption flagged and not tested

The split-attention agent framework gives an agent a finite attention budget and a rule for spending it (Sachikonye, 2026b). Theorem T2 establishes that the optimal spend equalises marginal return across activities—water-filling, the same rule that allocates power across parallel channels (Cover and Thomas, 2006) and, more generally, any separable resource under a concave objective (Boyd and Vandenberghe, 2004). Its hypothesis is that each activity’s return is concave in the effort spent on it.

That paper is explicit that concavity is an assumption about the *environment*, not a structural property of the allocator. It is stated, used, and—in the setting it was written for—plausible.

The honest thing to do with a flagged assumption is to measure it somewhere it matters.

The setting we measure it in is retrieval. A pipeline of the kind described by Doerr and Born (2025) invokes a reasoner at run time over records gathered from several endpoints. Before the reasoner can run, the records must be there; gathering them costs queries; the query budget is finite. This is an attention-allocation problem in the exact sense T2 addresses, with sources in place of activities and queries in place of attention. If T2 applies, the agent has a rule. If it does not, the agent needs to know that before it acts.

1.2 Why joins are the case that matters

Not all retrieval is alike, and the difference is not stylistic.

Retrieving *distinct entities* from a single endpoint is coupon-collecting: after e draws the expected coverage is $N(1 - (1 - 1/N)^e)$, which is concave everywhere (Feller, 1968; Motwani and Raghavan, 1995). T2 applies without qualification.

Retrieving *paginated* results returns a fixed number of new records per query until the source is exhausted. Marginal yield is constant, not decreasing, so T2’s hypothesis fails—but harmlessly, in the sense that a linear objective’s optimum is also at a corner and any allocation achieves the same total.

Retrieving a *federated join* is different in kind. A joined record exists only when *both* sides of the pair have been drawn. If each side is drawn independently with the same effort, the expected number of completed joins goes as the *square* of a coverage probability. This is the regime the pipelines in question actually operate in: Rhea joined to UniProt, ChEBI joined to Rhea (Bansal et al., 2022; Buil-Aranda et al., 2013; Hastings et al., 2016; The UniProt Consortium, 2023). It is also, as we show, the one regime of the three where T2’s failure changes what the agent should do.

1.3 What we prove

1. **The join yield surface is a sigmoid, and its inflection is exact.** $Y(e) = Np^2$ with $p = 1 - (1 - 1/N)^{e/2}$ is convex below $e^* =$

$2 \ln 2 / \ln(1/(1 - 1/N))$ and concave above, with $e^* \rightarrow 2N \ln 2$ and $Y(e^*) = N/4$ for every N (Theorem 2.3). The threshold needs one number the pipeline already has.

2. **The threshold reverses the optimal policy.** Below e^* the optimum is at a corner—not near one—and committing beats spreading by at least 17.9% over the sampled range and by 97% at the smallest budgets. Above e^* the ordering reverses (Theorem 3.1). A gate is therefore not a refinement; without one the allocator picks the worse of the two available actions, and does so most severely for the agents with the least budget.
3. **The state variable is normalised effort.** The optimum equalises e/e^* across joins, not e (Theorem 3.5). With corpora of 30 and 90 the optimum gives $2.00 e^*$ and $1.98 e^*$ while the raw efforts differ threefold.
4. **The allocator refuses.** Below every threshold no allocation rule available to it is well-founded, so it declines and reports the budget that would suffice (Theorems 4.2 and 4.4). We verify the refusal fires, that it does *not* fire above threshold, and that the budget it names actually resolves it.

1.4 What we do not claim

We do not claim T2 is wrong. Theorem 2.3 is a *bound*, and the negative control in Section 5 confirms the concave branch recovers T2 exactly: above e^* the optimum for two identical joins is the even split to within 0.0%. We do not claim the join model is the only model of federated retrieval—it assumes independent uniform draws on both sides, and Section 6 sets out what breaks when that fails. And we do not claim the allocator is optimal in general; we claim it agrees with brute force on the cases we checked, and we report the case where an earlier version did not.

2 The join yield surface

2.1 Model

Definition 2.1 (Federated join source). A *join source* is a pair of endpoints together with a set of N *joinable keys*: values of the shared attribute for which a record exists on both sides. N is

not the record count; it is the count of keys that would survive the join, which is obtainable from a count query before any retrieval is done.

Spending effort e on the source means issuing e queries, split evenly between the two sides, each returning one key drawn uniformly at random with replacement.

The replacement assumption is the conservative one: sampling without replacement would only make coverage rise faster, sharpening every conclusion below. Section 6 returns to this.

Lemma 2.2 (Yield). *Under Theorem 2.1, the expected number of completed joins after effort e is*

$$Y(e) = N p(e)^2, \quad p(e) = 1 - \left(1 - \frac{1}{N}\right)^{e/2}. \quad (1)$$

Proof. Each side receives $e/2$ draws. For a fixed key, the probability it is missed on one side after $e/2$ uniform draws with replacement is $(1 - 1/N)^{e/2}$, so it is present with probability $p(e)$. The two sides are drawn independently, so the key is joinable with probability p^2 . Summing the indicator over all N keys and using linearity of expectation gives Equation (1); note this requires no independence *between* keys. $\square$

Equation (1) is the whole paper in one line. p is concave. p^2 is not, because squaring is convex and increasing, and the composition of a convex increasing function with a concave one is neither in general. The next result says exactly where each behaviour holds.

2.2 The threshold

Theorem 2.3 (Convexity threshold and the quarter-corpus law). *Let $q = 1 - 1/N$ with $N \geq 2$. Then Y in Equation (1) satisfies*

- (i) $Y''(e) > 0$ for $e < e^*$ and $Y''(e) < 0$ for $e > e^*$, where

$$e^* = \frac{2 \ln 2}{\ln(1/q)}; \quad (2)$$

- (ii) $e^* \rightarrow 2N \ln 2$ as $N \rightarrow \infty$, from below;

- (iii) $Y(e^*) = N/4$ exactly, for every N .

Proof. Write $r(e) = q^{e/2}$, so $p = 1 - r$ and $r' = \frac{1}{2}r \ln q$. Then

$$Y' = 2Np(-r') = -Nrp \ln q,$$

and differentiating again,

$$Y'' = -N \ln q (r'p + rp') = -N \ln q \left(\frac{1}{2} r \ln q p + r \left(-\frac{1}{2} r \ln q \right) \right),$$

which simplifies to

$$Y''(e) = -\frac{N}{2} (\ln q)^2 r (p-r) = \frac{N}{2} (\ln q)^2 r (2r-1), \quad (3)$$

using $p-r = 1-2r$. Since $N > 0$, $(\ln q)^2 > 0$ and $r > 0$, the sign of Y'' is the sign of $2r-1$. Thus $Y'' > 0$ exactly when $r > \frac{1}{2}$, i.e. $p < \frac{1}{2}$, and $Y'' < 0$ exactly when $p > \frac{1}{2}$. Solving $r(e) = \frac{1}{2}$ gives $q^{e/2} = \frac{1}{2}$, hence $e = 2 \ln 2 / \ln(1/q)$, which is Equation (2). This proves (i).

For (ii), $\ln(1/q) = -\ln(1-1/N) = 1/N + 1/(2N^2) + O(N^{-3})$, so $e^* = 2N \ln 2 (1 + \frac{1}{2N} + O(N^{-2}))^{-1} = 2N \ln 2 - \ln 2 + O(N^{-1})$, approaching $2N \ln 2$ from below.

For (iii), at $e = e^*$ we have $r = \frac{1}{2}$ hence $p = \frac{1}{2}$, so $Y = Np^2 = N/4$ identically in N . $\square$

Remark 2.4 (Why (iii) is the useful part). Parts (i) and (ii) locate the threshold in units of effort, which an agent can compute but cannot easily interpret. Part (iii) states it in units of *result*: the threshold is precisely the point at which a quarter of the joinable corpus has been recovered. This holds exactly, for every N , with no asymptotics. An agent that has retrieved fewer than $N/4$ joins is below threshold; one that has retrieved more is above. The test requires no fitted constant and no calibration, and it can be evaluated on results already in hand rather than on a model of effort.

Corollary 2.5 (The convexity is structural). *The sign change in Equation (3) depends on N only through the positive factor $\frac{N}{2}(\ln q)^2$. No choice of corpus size, and no amount of sampling, removes the convex branch.*

Theorem 2.5 is stated because we needed it. Two earlier attempts to observe the convexity from simulation failed for reasons that had nothing to do with the physics (Section 5.3), and only evaluation of the closed form settled the matter.

3 The threshold reverses the optimal policy

That an objective is non-concave on part of its domain does not by itself mean an allocator

built for concave objectives is dangerous there. It might merely be suboptimal by a controllable amount. This section shows it is not: below the threshold the water-filling answer and the optimal answer are on opposite sides of the decision.

3.1 Corner versus interior

Theorem 3.1 (Policy reversal). *Consider two join sources with the same corpus size N and a shared budget B , allocated as $(x, B-x)$ with $x \in [0, B]$. Write $u = B/e^*$. There is a single crossover*

$$u^\dagger = 2 \log_2(1 + \sqrt{2}) = 2.5431 \dots, \quad (4)$$

the same for every N and not merely in the limit, such that:

- (a) *If $u < u^\dagger$, committing the entire budget to one source is optimal, and the optimum is the endpoint $x = 0$ (or $x = B$).*
- (b) *If $u > u^\dagger$, the symmetric split $x = B/2$ is optimal.*
- (c) *At $u = u^\dagger$ the two policies have exactly equal yield.*

In particular the optimum jumps from $x/B = 0$ to $x/B = 1/2$ discontinuously: no asymmetric interior split is ever optimal.

Proof. Work in normalised effort. Write $q = 1 - 1/N$ and $t = q^{B/2}$; since $q^{e^*/2} = \frac{1}{2}$ by the definition of e^* , we have exactly $t = 2^{-u/2}$ for every N , with no approximation. The commit policy then yields $N(1-t^2)^2$ and the even split $2N(1-t)^2$. Setting these equal,

$$(1-t)^2(1+t)^2 = 2(1-t)^2,$$

and since $t < 1$ we may divide by $(1-t)^2$ to get $(1+t)^2 = 2$, hence $t = \sqrt{2} - 1$ and $u = -2 \log_2(\sqrt{2} - 1) = 2 \log_2(1 + \sqrt{2})$, which is (4). This is the unique crossing, because $(1-t^2)^2 - 2(1-t)^2 = (1-t)^2[(1+t)^2 - 2]$ changes sign only where $(1+t)^2 = 2$; as t is strictly decreasing in u , the commit policy wins below $u^\dagger$ and loses above it.

For the stronger statement that these two are the only candidates: when $u < 1$ every feasible x leaves both sources on the convex branch, so the total is convex and attains its maximum at an extreme point (Boyd and Vandenberghe, 2004;

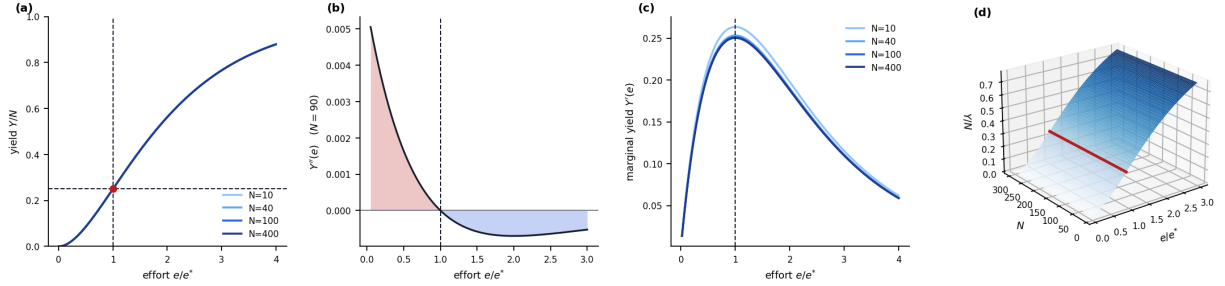


Figure 1: The Federated-Join Yield Surface and Its Inflection. (A) Normalised yield $Y(e)/N$ for corpus sizes $N \in \{10, 40, 100, 400\}$ against effort measured in units of each corpus’s own threshold e^* . The four curves are drawn from $Y(e) = Np^2$ with $p = 1 - (1 - 1/N)^{e/2}$; because p saturates and squaring a saturating function does not, every curve is sigmoid rather than concave. All four cross $Y/N = 0.25$ at exactly $e/e^* = 1$ (red marker), which is the quarter-corpus law: the yield at the inflection is $N/4$ for every N , with no asymptotics and no fitted constant. (B) Second derivative $Y''(e) = \frac{N}{2}(\ln q)^2 r(2r - 1)$, where $q = 1 - 1/N$ and $r = q^{e/2}$, evaluated at $N = 90$ ($e^* = 124.07$). The convex region $Y'' > 0$ is shaded red and the concave region $Y'' < 0$ blue. The sign change occurs at $e/e^* = 1$ and nowhere else, so the domain splits into exactly two branches: increasing returns below threshold, diminishing returns above it. (C) Marginal yield $Y'(e) = -Nrp \ln q$ for the same four corpora. Marginal yield does not decrease monotonically—it *rises* to a peak at $e/e^* = 1$ before falling. This peak is the mechanism behind the necessary-versus-sufficient failure: at a budget just above $\sum_i e_i^*$ every source sits barely onto its concave branch where its marginal return is still near maximal, so withdrawing one source outright can free enough budget to push another further up its curve. (D) Yield surface Y/N over $(e/e^*, N)$ for $N \in [10, 300]$, with the threshold ridge $e = e^*(N)$ traced in red at constant height 0.25. The ridge is flat in N : the inflection sits at the same normalised effort and the same fractional yield for every corpus size, which is why e/e^* and not e is the allocator’s state variable.

Horst, 1984). When $u \geq 2$ the total is concave on $x \in [e^*, B - e^*]$, so $x = B/2$ is the best *interior* point—but concavity there says nothing about $x \notin [e^*, B - e^*]$, and the endpoints lie outside it. The comparison above is what settles those, and it is why the crossover is at $u^\dagger$ rather than at the edge $u = 2$ of the concave region. $\square$

Remark 3.2 (Why the boundary is not $2e^*$). It is tempting to stop at the observation that both sources are on their concave branches once $B \geq 2e^*$, and conclude the symmetric split wins there. It does not. At $B = 2e^*$ exactly, committing yields $0.5625N$ against the even split’s $0.5N$ —the corner is still ahead by 12.5%, and stays ahead until $u = u^\dagger$. Concavity on the interior region is necessary for the interior optimum, not sufficient for it, because the competing optima sit on a part of the feasible set where the function is convex. This is the same necessary-versus-sufficient error that Theorem 4.5 records for the multi-source allocator; an earlier draft of this theorem asserted the $2e^*$ boundary, and Figure 2(a)—whose crossing is computed, not drawn—is what exposed it.

Remark 3.3 (A second exact constant). Substituting $t = \sqrt{2} - 1$ into either policy gives the common yield at the crossover in closed form,

$$Y(u^\dagger e^*) = 4(\sqrt{2}-1)^2 N = (12-8\sqrt{2})N \approx 0.68629N \quad (5)$$

exact for every N , as (4) is. It is the companion to the quarter-corpus law of Theorem 2.3(iii): e^* is where a join has recovered a quarter of its corpus, and $u^\dagger e^*$ is where it has recovered $12 - 8\sqrt{2}$ of it and an allocator’s optimal policy flips. Both are properties of the join surface alone, with no fitted constant and no dependence on corpus size.

The empirical content is in the magnitude. Over budgets from 0.15 to 0.90 of e^* , committing beats the even split at every budget by at least 17.9%; the worst case measured is a 97% relative loss at $0.02e^*$. The loss grows as the budget shrinks, so the agents with the least attention—the ones an allocation theory exists to help—are the ones an ungated allocator hurts most.

Remark 3.4 (The optimum is *at* a corner, not near one). Theorem 3.1(a) compares two policies, which leaves open that some third, interior policy beats both. Brute-force search over 2001 splits

of a budget of $0.50e^*$ places the unconstrained optimum at $x = 0$ exactly. There is no interior optimum being missed.

Sweeping the budget rather than fixing it makes the stronger point. Brute-force argmax over 4001 splits at $u \in \{1.0, 1.5, 2.0, 2.2, 2.4\}$ returns $x/B = 0.000$ at every one; at $u \in \{2.7, 3.0\}$ it returns $x/B = 0.500$. The optimal split never takes an intermediate value—it jumps at $u^\dagger$. An allocator therefore cannot approach the correct answer by tuning a split fraction continuously, which is a second reason the decision must be gated rather than smoothed.

3.2 Normalised effort

Identical corpora make the reversal easy to state and easy to under-generalise. The mixed case is where the allocation rule earns its keep.

Proposition 3.5 (The optimum equalises e/e^*). *For sources with corpus sizes N_i and thresholds e_i^* , at budgets where every source is on its concave branch, the optimal allocation equalises the normalised efforts $u_i = e_i/e_i^*$ to first order, rather than the raw efforts e_i .*

Argument. On the concave branch the optimum equalises marginal yields, $Y'_i(e_i) = \lambda$ for all i . From the proof of Theorem 2.3, $Y'_i = -N_i r_i p_i \ln q_i$ where $r_i = q_i^{e_i/2} = 2^{-u_i}$ by Equation (2), so r_i —and hence $p_i = 1 - r_i$ —depends on e_i *only* through u_i . Therefore $Y'_i(e_i) = N_i \ln(1/q_i) \cdot 2^{-u_i}(1 - 2^{-u_i})$. Since $N_i \ln(1/q_i) \rightarrow 1$ as $N_i \rightarrow \infty$ (from $\ln(1/q_i) = 1/N_i + O(N_i^{-2})$), the prefactor becomes common to all sources, and equalising λ reduces to equalising the function $u \mapsto 2^{-u}(1 - 2^{-u})$, hence to equalising u_i on the branch $u > 1$ where it is monotone. $\square$

Measured: with corpora of 30 and 90 ($e^* = 40.9$ and 124.1) and a budget of 327.1 , the optimum allocates 81.9 and 245.3 —that is, $2.00e^*$ and $1.98e^*$, equal to within 1.3%, while the raw efforts differ threefold. At three sources with corpora 20, 60, 200 the normalised efforts are 2.54, 2.51, 2.49, equal to within 1.7%, while raw efforts are 69, 207, 690.

Corollary 3.6 (A global threshold is wrong in both directions). *Because e^* depends on N , a single budget threshold applied to a mixed set of sources will gate a large-corpus join as though it*

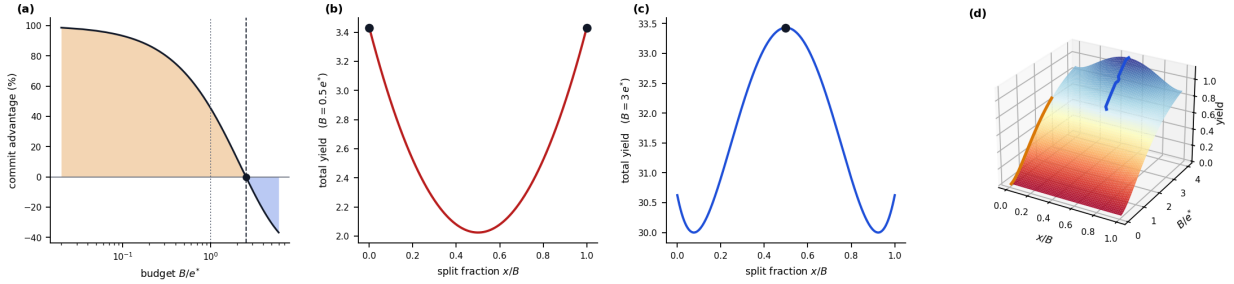


Figure 2: The Threshold Reverses Which Allocation Policy Is Optimal. (A) Relative advantage of committing the whole budget to one source over splitting it evenly, against budget B/e^* on a logarithmic axis, for two sources of equal corpus size $N = 40$. Committing wins by 98.6% at $B = 0.02e^*$, by 45.7% at $B = e^*$, and still by 12.5% at $B = 2e^*$ where both sources are already on their concave branches. The advantage crosses zero at $u^\dagger = 2 \log_2(1 + \sqrt{2}) = 2.5431$ (solid marker), *not* at the dotted line $B = e^*$ where the yield curve inflects. The crossing is located by the plotting code rather than drawn at a predicted value, and the gap between $u^\dagger$ and the edge of the concave region at $u = 2$ is what falsified an earlier version of the reversal theorem. (B) Total yield against split fraction x/B at a sub-threshold budget $B = 0.5e^*$. The curve is convex and attains its maximum at both endpoints (3.431 at $x/B \in \{0, 1\}$) against 2.025 at the centre—a 69.4% penalty for splitting. The optimum is *at* a corner, not near one. (C) The same construction at a super-threshold budget $B = 3e^*$. The ordering has reversed: the maximum is the symmetric split (33.43 at $x/B = 0.499$) against 30.63 at the endpoints. Water-filling is exactly right here and exactly wrong in (B). (D) Total yield over (split fraction, budget) for two equal corpora, with the ridge of optima traced—orange below the crossover, blue above. The ridge jumps from the boundary $x/B = 0$ to the centre $x/B = 1/2$ with no intermediate values, confirming the discontinuity: no asymmetric interior split is ever optimal, so an allocator cannot reach the answer by tuning a split fraction continuously.

were above threshold when it is below, and vice versa. Gating must be per source, in normalised units.

At the mixed-corpus optimum the small join receives 25.0% of the budget, yielding 66.971 against 58.659 for an even split (+14.2%) and 63.383 for committing everything to one source (+5.7%). The optimum is strictly interior and strictly better than both naive policies, so neither can imitate it: the rule has to be genuinely per-join.

4 The allocator

4.1 What the results dictate

The shape of the allocator is not chosen. Theorem 2.3 says each source carries its own threshold, computable from its corpus size. Theorem 3.1 says the optimal policy on either side of that threshold is different in kind. Theorem 3.5 says the state variable is $u = e/e^*$. Theorem 3.6 says the gate is per source. What remains is the case analysis.

Definition 4.1 (Gated allocation). Given sources $\{s_i\}$ with thresholds e_i^* and a budget B , let $F = \sum_i e_i^*$ and $m = \min_i e_i^*$.

- (1) If $B \geq F$: every source *can* be placed on its concave branch. Compute the water-filling solution, and compare it against every drop-one-source corner; return whichever yields more.
- (2) If $m \leq B < F$: some but not all sources can be lifted past threshold. Enumerate the subsets that the budget can lift, water-fill each, and return the best.
- (3) If $B < m$: **refuse**.

Definition 4.2 (Refusal to allocate). When $B < m$, no source can be brought onto its concave branch. Every allocation rule available is either founded on a hypothesis that does not hold (Y concave) or is a guess. The allocator therefore returns no allocation. It returns a *refusal*, carrying the offered budget, the required budget m , the shortfall, and the per-source thresholds.

This is the same move as the medium-vertex paper’s two refusals (Sachikonye, 2026a), and made for the same reason.

Principle 4.3 (Refusal, after Sachikonye, 2026a). A representation earns a positive result only if some presentable object would have received a negative one. A framework with no refusals has an empty defined class: it does not classify, it merely accepts.

The difference from that paper is worth noting. There, the refusal threshold is the individuation floor β , a parameter of the framework. Here, the threshold is e^* , derived from a count query against a public endpoint. The refusal is therefore not merely principled but *cheap*: an agent can decide whether it is entitled to allocate before it spends anything.

Theorem 4.4 (The refusal is non-vacuous). *There exist budgets at which Theorem 4.1 refuses and budgets at which it does not, and the refusal’s reported required budget suffices.*

Verification. For the pair ($N = 30, N = 90$) with $e^* = 40.89$ and 124.07: at $B = 20.4$ the allocator refuses, reporting a required budget of 40.89 and a shortfall of 20.45. At $B \in \{165, 247, 495, 1650\}$ it returns an allocation. Supplying exactly the reported 40.89 succeeds, returning a commit policy with yield 7.500. See Section 5, checks C.5–C.7. $\square$

The third clause matters as much as the first two. A gate that never closes is not a gate; a gate that always closes is not one either; and a refusal that misreports its own remedy is worse than no refusal, because it sends the caller to acquire a budget that will be declined again.

4.2 Interface

```
src = [JoinSource("chebi-rhea",
  corpus_size=30),
  JoinSource("rhea-uniprot",
  corpus_size=90)]

try:
  a = allocate(src, budget=300.0)
  a.per_source # {'chebi-rhea': 81.9,
  ...}
  a.normalised # {'chebi-rhea': 2.00,
  ...} <- e/e*
  a.policy # 'water-fill' | 'commit-
  subset'
except Refusal as r:
  r.required_budget # 40.89
  r.shortfall # what to acquire
```

Sources that receive nothing appear in `per_source` with value zero rather than being omitted. This is deliberate: a starved join is a question the pipeline did not answer, not one it answered negatively, and the caller must be able to tell those apart.

4.3 A necessary condition mistaken for a sufficient one

The comparison against drop-one-source corners in Theorem 4.1(1) is not defensive programming. It repairs a real error, and the error is instructive enough to state as a proposition.

Proposition 4.5 (Concave branch is necessary, not sufficient). *$B \geq \sum_i e_i^*$ places every source on its concave branch, and is therefore necessary for an interior optimum in which all sources are funded. It is not sufficient: there exist budgets $B \geq \sum_i e_i^*$ at which the optimum funds a proper subset.*

Proof. By construction. Take $N_1 = 30$, $N_2 = 90$, so $e_1^* = 40.89$, $e_2^* = 124.07$, $F = 164.96$. At $B = F$ exactly, the marginal-equalising interior solution allocates (e_1^*, e_2^*) , placing both sources precisely at threshold; by Theorem 2.3(iii) its yield is therefore $N_1/4 + N_2/4 = 30$ exactly. Funding only the larger source—allocating $(0, F)$ —yields 32.629, or 8.76% more. $\square$

Remark 4.6. The interior value in that proof is exactly the quarter-corpus law summed over sources, which is a useful sanity check on the construction: $B = \sum_i e_i^*$ is the unique budget at which the water-filling solution puts *every* source exactly at its inflection, and so recovers precisely a quarter of every corpus.

The mechanism is visible in Figure 1(c). Marginal yield Y' does not decrease monotonically; it *rises* to a peak at e^* and falls thereafter. At B just above F every source sits barely onto its concave branch, where its marginal yield is still near that peak. Withdrawing one source entirely then frees enough budget to push another far up its curve, and the gain outweighs the loss.

The first-order condition cannot detect this, because it compares interior points and the winner is a corner. This is the same structural fact as Theorem 3.1, appearing one level up: the KKT conditions are sufficient only under concavity of the objective *over the feasible set*, and

the feasible set here includes the faces where a source is unfunded (Boyd and Vandenberghe, 2004).

Remark 4.7 (Why a self-consistency check would have passed). The broken allocator satisfied its own optimality condition exactly: marginals were equalised to 10^{-10} . A test asserting “the returned allocation equalises marginal yield” would have passed. What caught it was checking against exhaustive search over the simplex—a procedure sharing no line of reasoning with the allocator. This generalises: an optimality check is only informative if it is not the optimality condition the implementation was built from.

Subset selection in Theorem 4.1(2) is likewise by enumeration rather than by a greedy ratio rule. A greedy rule was tried first and is not optimal, for the same reason: which subset wins depends on how far the budget pushes each survivor up its own curve, which is not a function of any per-source ratio computable before the split is known. Enumeration over subsets is affordable because a pipeline federates a handful of sources, not hundreds (Cormen et al., 2009).

5 Validation

5.1 What the suite can establish, and what it cannot

Every quantity in this paper is closed-form except where noted, so most checks are exact evaluations rather than estimates. That removes sampling error as an explanation for a pass, but it also means a passing check establishes only that the implementation computes what the formula says. Three things are done to make the suite capable of failing:

1. **Negative controls.** Six of the twenty-one checks assert that something does *not* happen on well-formed input: that the allocator does not refuse above threshold; that spreading does *not* lose above threshold; that the concave branch *does* recover T2.
2. **Independent cross-checks.** The allocator is compared against brute-force simplex search, which shares no reasoning with it (Theorem 4.7).
3. **A positive control on the detector.** Coupon-collector retrieval is analytically con-

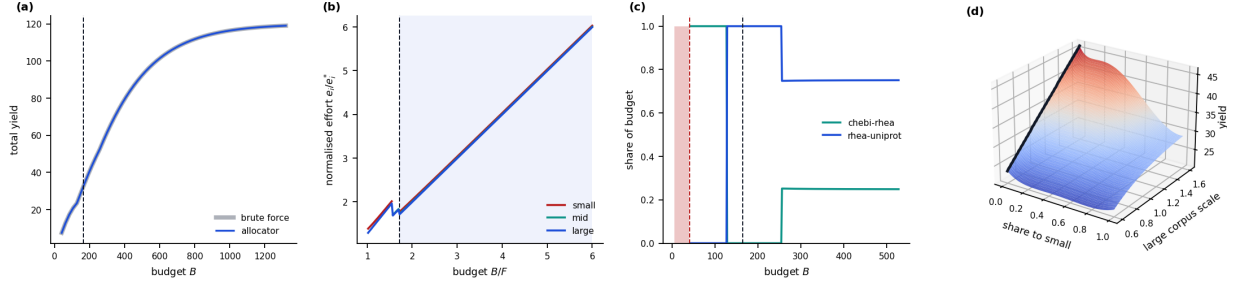


Figure 3: The Gated Allocator Against Exhaustive Search. (A) Allocator yield and brute-force yield against budget for the ChEBI–Rhea/Rhea–UniProt pair ($N = 30$ and 90 , so $e_1^* = 40.89$, $e_2^* = 124.07$, $F = \sum_i e_i^* = 164.96$). The two curves are visually indistinguishable across the full range; the worst relative gap over 60 budgets is 9.3×10^{-7} , which is the resolution of the 800-step brute-force grid rather than allocator error. The dashed line marks $B = F$. (B) Normalised efforts e_i/e_i^* against budget B/F for three unequal corpora $\{20, 60, 200\}$ ($e^* = 27.03, 82.48, 276.57$; $F = 386.07$). Within the shaded water-fill regime, which begins at $B/F = 1.72$, the three converge to a common value—at $B/F = 6$ they are $6.030, 6.004$ and 5.996 , agreeing to 0.03 despite raw efforts differing by an order of magnitude. To the left of the shaded band a source receiving nothing is drawn as a *gap* rather than as zero, because there it is unfunded by design under the commit-subset rule and not an equalisation that failed. (C) Share of budget allocated to each source across the three policy regimes. Below $B = e_{\min}^* = 40.89$ (red dashed) the allocator refuses outright, shaded in red. Between $e_{\min}^*$ and roughly $1.7F$ it commits to a proper subset. Only above that does it water-fill, at which point both shares stabilise. (D) Total yield over (share to the small source, scale factor on the large corpus) at a budget just above F , with the argmax ridge traced in black. The ridge sits pinned at share = 0 across the whole range of corpus scalings: at $B = F$ exactly the marginal-equalising interior solution places both sources precisely at threshold and yields $N_1/4 + N_2/4 = 30$, while funding only the larger source yields 32.629 — 8.76% more. Having every source on its concave branch is necessary for an interior optimum, not sufficient for it.

Experiment	Claim checked	Checks	Gain	Neg. Corr.
exp-a	Theorem 2.3 — the threshold	6	at three unequal sources with a 140-step search it is 1.4×10^{-6} (in the allocator's favour, i.e. within the search grid's resolution). Both figures are reported as measured rather than rounded to zero.	0.0000%
exp-b	Theorem 3.1, Theorem 3.5	7		
exp-c	Theorem 4.1, Theorem 4.4	8		
total		21	6	

Table 1: Validation summary. All 21 checks pass. Experiments run in dependency order: **exp-a** finds the threshold, **exp-b** shows it flips the optimal policy, **exp-c** tests the implementation built on both.

cave, so the diminishing-returns detector is run against it first. A failure there would indicate a broken detector rather than a finding about the world.

5.2 Results

Twenty-one checks across three experiments; all pass.

Four findings are worth stating beyond the pass count.

The three retrieval regimes separate cleanly. Distinct-entity retrieval shows marginal yield falling from 1.000 to 0.222 with 0/59 second differences exceeding tolerance—the positive control. Paginated retrieval holds marginal yield at 3.000 throughout (ratio 1.000), and is weakly concave with *all* second differences zero, which is why concavity alone is the wrong criterion and marginal *decay* is the right one. Federated joins show 31/59 second differences positive beyond tolerance.

The convexity is structural, not statistical. The closed form evaluated with no sampling is convex at 54/59 second differences (maximum +0.01234). No number of trials would have made the sampled curve concave. This check exists because two earlier failures made the finding look like an artefact (Section 5.3).

The quarter-corpus law holds across two decades of N . $e^*/(2N \ln 2)$ rises from 0.949 to 0.999 over $N = 10 \dots 1000$, and $Y(e^*) = N/4$ at every N tested, to machine precision.

The allocator matches brute force. Over budgets from F to $8F$ the worst relative gap

5.3 Two measurement errors and two real ones

The medium-vertex paper reports a failing check that became Proposition 7.1 (Sachikonye, 2026a). The same discipline applies here, and four episodes are worth recording because each changed what we believed.

Monte Carlo noise five times the signal.

The first attempt to observe convexity sampled the join process directly. At the trial count used, the sampling standard error on each second difference exceeded the true curvature by roughly a factor of five, and the arm failed. The correct response was not more trials—which we did try—but the realisation that the question was answerable exactly.

A parity artefact. The second attempt split effort between the two sides using an alternating rule indexed by the sample number. This introduced a period-two component into the yield curve, and the second-difference detector reported convexity at alternating indices $[0, 2, 4, 6, 8, 9]$, a pattern with no physical meaning. A diagnostic comparing parity, random and analytic splitters showed the analytic curve—zero noise, no splitter at all—convex at 54/59 points. That settled it, and Theorem 2.5 is the statement we should have started from.

The allocator flaw. Theorem 4.5 was not anticipated. The allocator was written with $B \geq F$ as the water-filling gate, which is the natural reading of Theorem 2.3 and is wrong. It was caught by a single check—agreement with brute force—failing at 8.0566% at one budget while agreeing to 0.000% at every other. That the discrepancy was budget-dependent is what made it diagnostic rather than merely alarming.

A theorem with the wrong boundary, caught by its own figure. Theorem 3.1 originally asserted that the symmetric split is optimal

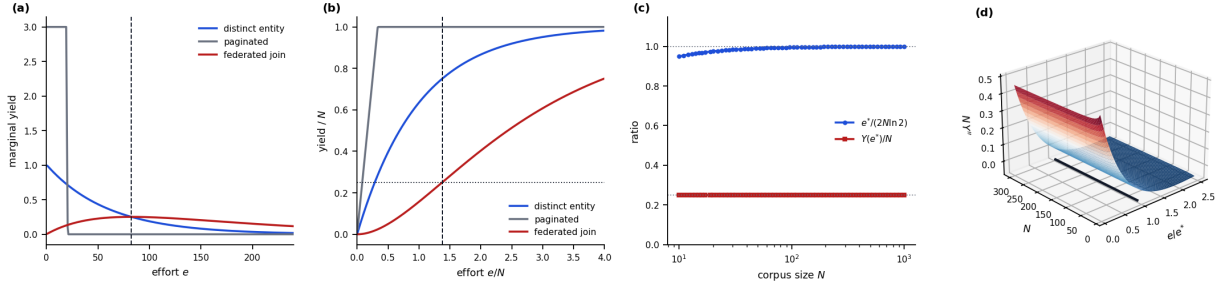


Figure 4: **Three Retrieval Regimes, and Two Constants Independent of Corpus Size.**

(A) Marginal yield against effort for three retrieval regimes on common axes at $N = 60$ ($e^* = 82.48$, dashed). Distinct-entity retrieval follows coupon-collector coverage $N(1 - (1 - 1/N)^e)$ and falls monotonically. Paginated retrieval is flat at the page size until the corpus is exhausted, then zero. The federated join alone *rises* to a peak at e^* before falling. Only the join violates diminishing returns, and it does so across the entire sub-threshold range—so the assumption underpinning water-filling is environmental rather than structural, and it fails in precisely the environment a federated pipeline operates in. (B) The same three regimes as cumulative normalised yield Y/N against e/N . The dotted line at 0.25 and the dashed line at $e^*/N = 1.375$ intersect on the join curve, which is the quarter-corpus law located on the same axes as its two controls. (C) Two ratios against corpus size on a logarithmic axis, N from 10 to 10^3 . The asymptotic form of the threshold, $e^*/(2N \ln 2)$, approaches 1 from below but is only 0.949 at $N = 10$ and reaches 0.9995 at $N = 1000$ —an approximation that improves with N . The quarter-corpus law $Y(e^*)/N$ is pinned at 0.25 to within 3.0×10^{-14} at every N tested. One constant is asymptotic; the other is exact. (D) Scaled second derivative NY'' over $(e/e^*, N)$, red where convex and blue where concave, with the zero contour projected onto the floor. The contour is the plane $e = e^*$, flat in N : the boundary between increasing and diminishing returns sits at the same normalised effort for every corpus size, which is the threshold theorem rendered as a surface.

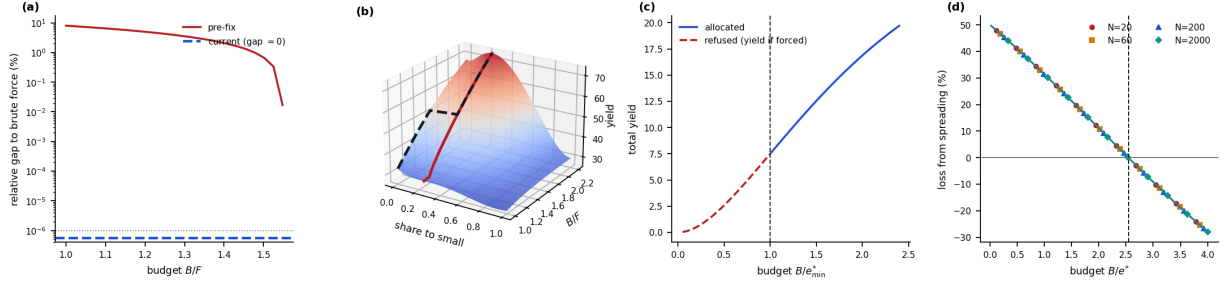


Figure 5: An Allocator Flaw Caught by Brute Force, and the Refusal. (A) Relative shortfall against exhaustive search, on a logarithmic axis, for the pre-fix allocation rule (water-fill whenever $B \geq F$) and for the shipped gated rule, over 220 budgets from F to $6F$. The pre-fix rule is suboptimal at 25 of them, forfeiting up to 8.0566% of the achievable yield at $B = F$ and recovering only past $B = 1.548F$. The gated allocator’s shortfall is identically zero at every budget, so it has no curve to plot and is marked on the floor instead. A self-consistency check would have passed the broken rule; only comparison against a search sharing none of its reasoning exposed it. (B) The two allocations traced across the yield surface for $B/F \in [1.0, 2.2]$. The pre-fix interior solution (solid red) sits below the corner the gated rule selects (dashed black) precisely over the region where the surface is still convex, and the two merge once the budget is large enough that the interior solution is genuinely optimal. (C) Total yield against budget in units of $e_{\min}^* = 40.89$. Below $e_{\min}^*$ the allocator refuses to answer; the dashed red continuation is the yield it would have reported had it been forced to. The refusal is not a failure to compute—the number exists—but a statement that no allocation over these sources at this budget is defensible, computable from corpus size alone. (D) Loss from spreading rather than committing, against B/e^* , for four corpus sizes spanning two orders of magnitude ($N = 20, 60, 200, 2000$; markers offset along each curve so the overlap is visible). The four curves coincide to 1.3×10^{-11} percentage points: in normalised effort N cancels exactly, so the penalty for ignoring the gate is a function of B/e^* alone. The loss runs from 49.7% at $B = 0.02 e^*$ through zero at $u^\dagger = 2.5431$ (dashed) to -28.0% at $B = 4e^*$, where spreading has become the better policy by the same margin.

once $B \geq 2e^*$, reasoning that both sources are then on their concave branches. Drawing Figure 2(a)—which computes the crossover rather than marking a predicted one—put the zero crossing at 2.5431, not at the 2 the theorem claimed. The theorem was wrong: at $B = 2e^*$ exactly, committing beats the even split by 12.5%, and the true boundary is the N -independent constant $u^\dagger = 2\log_2(1 + \sqrt{2})$ of (4).

The error is instructive because it is the *same* error as Theorem 4.5, made a second time in a different place. Concavity on the region where all sources exceed threshold is necessary for the interior optimum and not sufficient for it, because the competing corner solutions lie outside that region. Having identified that trap for the multi-source allocator, we reproduced it in the two-source theorem, and the proof passed inspection because every individual step in it is true—it simply never compared the interior stationary point against the endpoints. What caught it was insisting that figures compute their own reference lines instead of drawing the value the text predicts.

We also record a check that *passed vacuously* and had to be replaced. An early mixed-corpus test used corpora of 12 and 300, where the optimum happens to lie at 100% to the small join and therefore coincides with the commit policy. It passed, but it could not distinguish a genuine per-join rule from “always commit”. A sweep over corpus pairs found 30/90 at budget 327.13, where the optimum is strictly interior and strictly beats both naive policies—the case now reported in Section 3. A check that cannot discriminate between the hypothesis and its competitor is not evidence for it, whatever its verdict says.

6 Scope

Theorem 2.1 makes four assumptions, and each bounds the result differently.

Uniform draws. Real endpoints return results in an order correlated with the query, not uniformly at random. Non-uniformity concentrates coverage on a subpopulation, which raises p early and lowers it late—sharpening the sigmoid rather than removing it. The threshold moves; the convex branch does not disappear.

With replacement. Sampling without replacement makes coverage rise strictly faster, reaching $p = 1/2$ at lower effort. This lowers e^* and therefore *shrinks* the refusal region. Our threshold is thus conservative in the direction that matters: it refuses slightly more often than necessary rather than allocating where it should not.

Independent sides. If presence on one side correlates with presence on the other—plausible when both endpoints derive from a common curation effort (Bansal et al., 2022)—then $\Pr[\text{both}] > p^2$ and yield is larger than Equation (1). In the limit of perfect correlation, yield reduces to Np , which is concave everywhere and T2 applies unconditionally. The convex branch is therefore a consequence of *independence*, and an agent that knows its sources are correlated should say so.

Even split between sides. Effort is split evenly. An agent free to allocate asymmetrically *between* the two sides of one join has a further optimisation available which we have not characterised. We expect it to matter most when the two sides have very different N —which Theorem 2.1 excludes by construction, since N counts keys joinable on both—but we have not measured it.

Two further limits are worth naming. We have not connected the allocator to the coordination machinery of T1–T7: an agent that refuses one join should presumably re-offer that budget to its peers, and the conserved character invariant and Kuramoto coupling (Kuramoto, 1975; Sachikonye, 2026b) are the natural mechanism, but we have not built it. And we have not run the allocator against live endpoints; every number here is closed-form or simulated, and a corpus size estimated from a count query carries its own error which we have not propagated.

7 Conclusion

An assumption that a theorem flags as environmental should be measured in the environment. We measured T2’s in federated retrieval and found it false below an effort threshold that is exact, closed-form, and computable from a count query: $e^* = 2\ln 2 / \ln(1/(1 - 1/N)) \rightarrow 2N \ln 2$,

equivalently the point at which a quarter of the joinable corpus has been recovered. The failure is not a matter of degree. Below the threshold the optimum is at a corner and water-filling recommends the opposite action, losing up to 97% of achievable yield, worst for the agents with the least budget.

What follows is not a repair of T2 but a gate on it. The allocator works in normalised effort e/e^* , gates per source rather than globally, water-fills where the hypothesis holds, commits where it does not, and refuses where neither is founded—reporting the budget at which it could proceed. This is the same shape of refusal the medium vertex supplies to the receiver language (Sachikonye, 2026a), arrived at independently and for the same reason: a system that always returns an answer has not said anything by returning one.

The methodological residue may be the more portable part. Two of our three surprises were our own measurement errors, and both initially looked like findings; the third was a genuine flaw that satisfied the implementation’s own optimality condition exactly and was caught only by a check that shared no reasoning with it. A suite of checks that an implementation would pass by construction measures the implementation’s self-consistency, which was never in doubt.

Reproducibility

All experiments are in `models/publications/attention-allocated-retrieval/validation/`. Running `python run_all.py` executes `exp-a`, `exp-b` and `exp-c` in dependency order, propagates exit codes, and writes per-experiment JSON together with an aggregate to `results/`. Every numerical value quoted in this paper is drawn from those files. The allocator is `validation/kernel/allocator.py`; the brute-force comparison it is checked against is in `exp_c_allocator.py` and is deliberately written without reference to it.

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